

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0483

Lourich 138 kV
November 17, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0483. The Interconnection Customer proposes a 73.6 MW solar photovoltaic facility interconnecting at the Lourich 138 kV substation in the Pennsylvania area. The request was received on June 02, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 238 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,770,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Altavista Power Sta Substation to Rbidoux 138 kV, the W Kelso to Hutchinson Plant 1 138 kV loads to 117.0% of its 469 MVA rating. Project DFAX: 8.3%.

For the loss of Altavista Power Sta to Rubidoux 138 kV Line, the West Kelso to Dry Gap 138 kV loads to 116.2% of its 482 MVA rating. Project DFAX: 8.3%.

The Four Hills to Rubidoux 138 kV is overloaded to 109.2% of its 430 MVA emergency rating for the loss of the West Kelso to Dry Gap 138 kV line. The project contributes 6.0% DFAX (4.4 MW) to this overload.

The Altavista Power Statin Substation to Rubidoux 138 kV is overloaded to 107.9% of its 458 MVA emergency rating for the loss of the W Kelso to Dry Gap 138 kV. The project contributes 6.0% DFAX (4.4 MW) to this overload.

Hutchinson Plant 1 to Watford City 138 kV Line: 107.7% loading (488 MVA) following the outage of the Yacht Club D.S to Hutchinson Plant 1 Substation 138 kV line; DFAX = 6.9%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W Kelso to Hutchinson Plant 1 138 kV	Rebuild line with higher-capacity conductor	\$9,969,000	9% (\$901,000)
West Kelso to Dry Gap 138 kV	Rebuild line with higher-capacity conductor	\$4,065,000	9% (\$367,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Four Hills to Rubidoux 138 kV	Reconductor line	\$4,694,000	8% (\$369,000)
Altavista Power Statin Substation to Rubidoux 138 kV	Reconductor line	\$896,000	8% (\$70,000)
Hutchinson Plant 1 to Watford City 138 kV Line	Reconductor line	\$720,000	9% (\$63,000)
	Total allocated to project		\$1,770,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.